

Roll No.

TBT-801

B. TECH. (BIOTECHNOLOGY) (EIGHTH SEMESTER) END SEMESTER EXAMINATION, May, 2023

OMICS TECHNOLOGIES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Give a critical account on Solexa (Illumina) methods of genome sequencing. (CO1)
- (b) Write a short note on human genome components and its organization. (CO1)
- (c) Write short note on the following : (CO1)
 - (i) Exon-Intron Trapping
 - (ii) Sanger's method
2. (a) What is random mutagenesis ? Write a brief account of tagging mutagenesis. (CO2)

P. T. O.

(2)

- (b) Write short notes on the following :
- (i) Knock-out mouse
 - (ii) Gene deletion (CO2)
 - (c) Explain working principle of Anti-Sense Technology. (CO2)
3. (a) Illustrate the terms Epigenetics and Epigenomics and their applications. (CO3)
- (b) Comment upon importance of model organisms in omics technology. Explain Yeast as model organisms. (CO3)
- (c) Describe the salient features of *Arabidopsis thaliana* as a model organism in context to development of stress resistant varieties. (CO3)
4. (a) Explain the Yeast two-hybrid system in proteomics. (CO4)
- (b) Write a note on principal steps of 2D- SDS PAGE analysis. (CO4)
- (c) Illustrate working principle of mass spectroscopy and its application. (CO4)
5. (a) Differentiate between RT-PCR and conventional PCR. (CO5)
- (b) Explain the working of DNA microarray with the help of any *one* example. Where is it utilized ? (CO5)
- (c) Discuss term transcriptomics and its various application. (CO5)

Roll No.

TBT-802

**B. TECH. (BIOTECH.) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

ENTREPRENEURSHIP DEVELOPMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define entrepreneurship. Differentiate between Entrepreneur and Manager. (CO1)
- (b) What is Small Scale Industries ? Describe the role of small scale industries in the Indian Economy. (CO2)
- (c) Write minimum *five* factors influencing Entrepreneurship in India. (CO1)
2. (a) What are the different forms of business ownership ? Define with example. (CO2)
- (b) Discuss the different stages of entrepreneurship process with example. (CO3)
- (c) What is NDA ? Why is it very essential part of any business startup ? (CO3)

P. T. O.

(2)

3. (a) What is the role of government in entrepreneurship ? To what extent should it help protect people from entrepreneurship or should it simply get out of the way and leave the market to reward or punish inappropriate behavior ? (CO5)
- (b) Think honestly about your entrepreneurial intentions. Make a list of problems that you are likely to face by becoming an entrepreneur. Make a list of what you to gain by becoming an entrepreneur. (CO6)
- (c) Explain the term business incubator and its functions. (CO4)
4. (a) Define Franchising. Describe different forms of franchise with example. (CO5)
- (b) "Development requires resources but resources alone don't produce development." Examine functions of entrepreneur in the light of this statement. (CO2, CO3)
- (c) Explain Rural, Social and Women Entrepreneurship with suitable example.
5. (a) Write in detail about role of Marketing in Entrepreneurship Development. (CO2)
- (b) "Today we see women doing well in the field of entrepreneurship, but still not many women are motivated to become entrepreneurs." Explain the constrains restricting the growth of women entrepreneurship in India. (CO4, CO6)
- (c) List the various sources of funds for social enterprise. (CO5)